Reg. No:……

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University of Vavuniya

IT 3232 – Ecommerce(P)

Assignment 01

Time: 60 minutes **Date: 12nd March 2026**

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01. Develop a **Spring Boot REST API** for a simple **eCommerce Product Management System**. The system should allow administrators to **add, view, and delete products**.

You must implement the following layers:

- Model
- Service
- Controller

Use an **ArrayList** to store product data temporarily.

1. Product Structure

Create a Product class with the following attributes:

Attribute	Data Type
id	**int**
name	String
price	double
quantity	int

a) Add Product (POST)

Create an API to add a new product. Example request:

```
POST /products
```

The API should return a message "The product is Successfully Added in the list.". Example JSON body:

```
{
"id": 1,
"name": "Laptop",
"price": 250000,
"quantity": 5
}
```

Condition

When adding a product:

- **Quantity must be greater than 0**
- If quantity is **0 or negative**, the system should return an error message.

Example message:

```
Quantity must be greater than zero
```

b) Get All Products (GET)

Create an API to retrieve all products in the system.

```
GET /products
```

The API should return a list of products.

c) Delete Product by Name (DELETE)

Create an API to delete a specific product using the product id.

Example:

```
DELETE /products/1
```

The API should return a message "Product id 1 is successfully deleted.".

d) Delete All Products (DELETE)

Create an API to delete all products.

```
DELETE /products/all
```

The API should return a message "All Products successfully deleted.".